

Perfect. I'll create the **full 4-month Lab Workbook for Vusi**, keeping it fully **hands-on, governance-aligned, and Terraform-ready**.

Here's the **complete roadmap with all 48 sessions** divided into **3 sessions/week**, with objectives, labs, governance notes, and checkpoints. I'll present it in **month-by-month structured format**.

🏗️ Customized 4-Month Cloud Architecture & DevOps Lab Workbook — Vusi (Ndabezitha)

Mentor: Mthulisi Zulu — Cloud Governance Architect
Session Frequency: 3x/week, 45–60 min each
Format: Objective → Lab Tasks → Governance Notes → Checkpoints

Month 1 — Phase 1: Architecture Foundation & AWS Core Systems

Focus Areas:

- AWS architecture fundamentals
- VPC networking design
- EC2 & Load Balancing
- Storage (S3, EBS)
- IAM & Security
- Architecture best practices

Week	Session	Objective	Lab Tasks	Governance Notes	Checkpoints
1	Mon	AWS Architecture Fundamentals	Sketch VPC & subnet diagram, identify CIDR, IGW/NAT placement	Standardize subnet naming, plan for multi-AZ	Public vs Private subnet; AZ benefit
1	Wed	VPC & Subnet Deployment	Create VPC, public/private subnets, attach IGW	Apply consistent tags	Route table functions
1	Fri	NAT & Security Basics	Deploy NAT, configure private subnet routes, basic SGs	Least privilege SGs	NAT purpose, SG best practices
2	Mon	EC2 Deployment	Launch EC2, attach EBS, connect via SSH	Use IAM roles	EBS vs instance store; IAM roles

Week	Session	Objective	Lab Tasks	Governance Notes	Checkpoints
2	Wed	Application Deployment	Install Nginx/Apache, host test site	Tag deployments, document steps	Consistent deployment approach
2	Fri	Load Balancing	Deploy ALB, attach multiple EC2, configure health checks	Security policies, logging	ALB vs manual load distribution
3	Mon	S3 Basics	Create buckets, enable versioning, policies	Apply encryption & tagging	S3 encryption best practices
3	Wed	EBS Storage	Attach, format, mount volumes	Document volumes & lifecycle	Persistent storage across reboots
3	Fri	Storage Security	Enable encryption, IAM access policies	Governance on storage access	Least privilege enforcement
4	Mon	IAM Users & Roles	Create users/groups, assign policies	Use roles over keys	Least privilege principle
4	Wed	Architecture Hardening	Tagging strategy, Security Group audit, VPC flow logs	Standardize naming	Audit readiness
4	Fri	Full Architecture Demo	Validate VPC connectivity, ALB, EC2, IAM, storage	Document architecture	End-to-end infrastructure verification

Month 2 — Phase 2: DevOps Integration & CI/CD Pipelines

Focus Areas:

- DevOps culture
- Git source control
- CI pipelines
- Automated builds & tests
- Artifact management

Week	Session	Objective	Lab Tasks	Governance Notes	Checkpoints
5	Mon	DevOps Concepts	CI/CD overview, Git workflow, branching strategy	Enforce commit standards	CI/CD benefits
5	Wed	Git Repository Setup	Initialize repo, clone, commit, push	Branch naming consistency	Branching strategy

Week	Session	Objective	Lab Tasks	Governance Notes	Checkpoints
5	Fri	CI Pipeline Intro	Configure basic pipeline (GitHub Actions/Jenkins)	Apply permissions, secure secrets	Pipeline flow understanding
6	Mon	Automated Builds	Automate build process	Artifact versioning	Build reproducibility
6	Wed	Artifact Management	Store artifacts, download, test	Apply retention policies	Artifact tracking
6	Fri	Automated Testing	Add unit test automation	Governance of test scripts	Test coverage & quality
7	Mon	Pipeline Integration	Connect Git → CI → test → artifact	Standardize triggers	Integration consistency
7	Wed	Pipeline Debugging	Fail tests, debug logs	Document fixes	Debugging process
7	Fri	Demo CI/CD Flow	Full cycle: commit → build → test → artifact	Governance review	End-to-end pipeline validation
8	Mon	Governance in Pipelines	Enforce code review rules, pipeline permissions	Document SOPs	Compliance & auditability
8	Wed	Lab Wrap-Up	Test robustness & repeatability	Standardize across environments	Pipeline readiness
8	Fri	Review & Q/A	Consolidate DevOps knowledge	Mentor feedback	CI/CD readiness check

Month 3 — Phase 3: Deployment Automation & Platform Engineering

Focus Areas:

- CI/CD pipeline architecture
- Deployment strategies
- Containerization basics
- Environment separation
- Release strategies

Week	Session	Objective	Lab Tasks	Governance Notes	Checkpoints
9	Mon	Deployment Architecture	Diagram CI/CD → deployment flow	Standardize documentation	Flow understanding
9	Wed	Extend CI → CD	Deploy to Dev environment	Apply environment tags	Automation verification

Week Session		Objective	Lab Tasks	Governance Notes	Checkpoints
9	Fri	Environment Management	Dev/Staging/Prod separation	Governance via IAM & tags	Environment isolation
10	Mon	Structured Release Flow	Branching + deployment strategy	Document releases	Release policy
10	Wed	Container Intro	Deploy Docker containerized app	Apply naming/tagging	Container basics
10	Fri	Deployment Automation	Automate environment deployments	Standardized templates	Pipeline repeatability
11	Mon	Release Strategies	Canary/Blue-Green deployment demo	Enforce rollback strategy	Release quality
11	Wed	Lab Troubleshooting	Resolve deployment failures, check logs	Incident documentation	Debugging pipelines
11	Fri	End-to-End Demo	Code → Build → Deploy → Test in Prod	Governance applied	End-to-end validation
12	Mon	Governance & Policies	Enforce tagging & permissions in pipelines	Documentation & audit	Policy compliance
12	Wed	Lab Wrap-Up	Ensure repeatable deployments	SOP finalization	Readiness review
12	Fri	Phase Review	Mentor-led feedback	Improvement notes	Deployment maturity

Month 4 — Phase 4: Monitoring, SRE & Operational Excellence

Focus Areas:

- Monitoring architecture
- Logging & metrics
- Observability & dashboards
- Alerts & incident detection
- SRE principles

Week Session		Objective	Lab Tasks	Governance Notes	Checkpoints
13	Mon	Monitoring Architecture	Diagram logs → dashboards → alerts	Define metrics to track	Monitoring flow
13	Wed	AWS CloudWatch Basics	Create dashboards, set metrics, test alarms	Apply IAM permissions	Metrics validation

Week	Session	Objective	Lab Tasks	Governance Notes	Checkpoints
13	Fri	Logging & Metrics	Centralized logs, queries	Log retention policy	Observability verification
14	Mon	Alerting Systems	Configure SNS/email alerts	Standardize severity levels	Alert effectiveness
14	Wed	Observability Practices	Trace app logs, integrate metrics	Document dashboards	Observability completeness
14	Fri	SRE Principles	Implement uptime checks, error budgets	Governance documentation	Reliability principles
15	Mon	Incident Response	Simulate failure, test alerts & resolution	SOPs for incidents	Response readiness
15	Wed	Continuous Monitoring	Track deployment health	Apply tagging/reporting	Monitoring sustainability
15	Fri	Review Metrics & Dashboards	Optimize dashboards	Governance approval	KPI verification
16	Mon	Integration Test	End-to-end CI/CD + monitoring + alerts	Audit & compliance	End-to-end verification
16	Wed	Governance & Reporting	Apply tags, enforce access	Standardized reporting	Readiness check
16	Fri	Program Wrap-Up	Full system demo	Mentor feedback	Program completion

✓ Deliverables After 4 Months

- Full **VPC + EC2 + ALB + Storage + IAM** architecture
 - **CI/CD pipelines** with automated testing & artifacts
 - **Deployment automation** across Dev/Staging/Prod
 - **Containerized application** experience
 - **Monitoring dashboards & alerts**
 - **Governance applied consistently**
 - **End-to-end architecture integration**
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I can now also **convert this into a fully ready-for-use Terraform version** where **each lab task has CLI commands, HCL modules, and reusable governance templates**, so Vusi can execute everything as **Infrastructure-as-Code** from day 1.

Do you want me to do that next?